

A candidate proof of Three-chromatic (0,2)-graphs

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

Every finite 3-colourable (0,2)-graph is bipartite, so none has chromatic number exactly three. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Three-chromatic \(0,2\)-graphs](http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs) from http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs.

1.1 Problem statement

Question Are there any (0,2)-graphs with chromatic number exactly three?

1.2 Catalog context

A (0,2)-graph is a graph such that every pair of distinct vertices has either 0 or 2 common neighbours. It is fairly easy to see that a (0,2)-graph is necessarily regular and a variety of other properties can be shown to hold. Although (0,2)-graphs with chromatic number 2, 4 and 5 are known it is open as to whether there can be any (0,2)-graphs with chromatic number exactly three.

1.3 Claimed result

Model verdict: proved (confidence: high).

Every finite 3-colourable (0,2)-graph is bipartite, so none has chromatic number exactly three.

The model's own caveat: For finite simple undirected graphs; literature priority has not been independently checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 Statement

Theorem. Let G be a finite simple graph in which every pair of distinct vertices has either zero or two common neighbours. If G admits a proper colouring with three colours, then G is bipartite.

The proof uses a neighbourhood-matching property to show that every real-valued edge function whose sum around each quadrangle is zero is a vertex-potential difference. A proper three-colouring supplies precisely such an edge function.

2.2 1. The neighbourhood-matching property

For an edge uv , put

$$A = N(u) \setminus \{v\}, \quad B = N(v) \setminus \{u\}.$$

For each $a \in A$, the vertices v, a have the common neighbour u , and hence have a unique second common neighbour b . Define $\phi(a) = b$. Then $b \in B$, $ab \in E(G)$, and

$$u, v, b, a, u$$

is a quadrangle with four distinct vertices.

This map is a bijection. Indeed, u, b have v, a as their two common neighbours. Consequently, the analogous construction from B to A —taking the common neighbour of u, b other than v —is its inverse.

Thus, for every edge uv , there is a bijection

$$\phi : N(u) \setminus \{v\} \longrightarrow N(v) \setminus \{u\} \quad \text{with} \quad a\phi(a) \in E(G). \quad (1)$$

In particular, adjacent vertices have equal degrees, so each connected component is regular. The matching property itself, rather than regularity, is what we need.

2.3 2. A lemma about edge functions

An **antisymmetric edge function** assigns a real number $\omega(x, y)$ to every ordered adjacent pair, with

$$\omega(y, x) = -\omega(x, y).$$

Say it is **square-closed** if its sum around every quadrangle is zero.

Lemma. Suppose a finite connected graph satisfies (1). Every square-closed antisymmetric edge function ω has the form

$$\omega(x, y) = h(y) - h(x) \quad (2)$$

for some $h : V(G) \rightarrow \mathbb{R}$.

Proof. Define

$$\operatorname{div} \omega(x) = \sum_{y \in N(x)} \omega(x, y), \quad \Delta h(x) = \sum_{y \in N(x)} (h(y) - h(x)).$$

Antisymmetry gives

$$\sum_x \operatorname{div} \omega(x) = 0.$$

The Laplacian of a finite connected graph has image equal to the space of zero-sum functions. For completeness, it is symmetric, and

$$\langle h, -\Delta h \rangle = \sum_{\{x, y\} \in E(G)} (h(x) - h(y))^2$$

shows that its kernel consists exactly of the constants. Hence there exists h satisfying

$$\Delta h = \operatorname{div} \omega.$$

Set

$$\eta(x, y) = \omega(x, y) - (h(y) - h(x)).$$

Then η is antisymmetric, square-closed, and divergence-free:

$$\sum_{y \in N(x)} \eta(x, y) = 0 \quad \text{for every } x. \quad (3)$$

Suppose η is not identically zero. By finiteness and antisymmetry, its maximum over ordered edges is some $M > 0$. Choose an ordered edge uv with

$$\eta(u, v) = M.$$

Use the matching (1). For $a \in A$, write $b = \phi(a)$. Square-closedness on u, v, b, a, u gives

$$\eta(u, v) + \eta(v, b) - \eta(a, b) - \eta(u, a) = 0.$$

Therefore

$$\eta(v, b) - \eta(u, a) = \eta(a, b) - M \leq 0.$$

Summing over the matching yields

$$\sum_{b \in B} \eta(v, b) \leq \sum_{a \in A} \eta(u, a). \quad (4)$$

But (3) and antisymmetry identify the two sides as

$$\sum_{b \in B} \eta(v, b) = -\eta(v, u) = M, \quad \sum_{a \in A} \eta(u, a) = -\eta(u, v) = -M.$$

Thus (4) says $M \leq -M$, contradicting $M > 0$.

Consequently $\eta = 0$, proving (2). $\square$

2.4 3. Applying the lemma to a three-colouring

Work in a connected component of G ; a one-vertex component is harmless. Let

$$c : V(G) \rightarrow \mathbb{Z}/3\mathbb{Z}$$

be a proper colouring. For each ordered edge xy , define $\omega(x, y)$ to be the unique number in $\{-1, 1\}$ satisfying

$$\omega(x, y) \equiv c(y) - c(x) \pmod{3}. \quad (5)$$

This is well-defined and antisymmetric.

Consider any quadrangle. The sum of ω around it is:

- divisible by 3, by telescoping (5); and
- an even integer between -4 and 4 , being a sum of four numbers from $\{-1, 1\}$.

The only possibility is zero. Thus ω is square-closed.

The lemma now provides $h : V(G) \rightarrow \mathbb{R}$ such that, on every edge,

$$h(y) - h(x) = \omega(x, y) \in \{-1, 1\}. \quad (6)$$

Subtract a constant so that $h(r) = 0$ at one chosen vertex. Connectivity and (6) then imply that every $h(x)$ is an integer. Moreover, adjacent vertices have opposite parities. Hence

$$x \mapsto h(x) \pmod{2}$$

is a proper two-colouring.

Applying this independently to every component proves that G is bipartite. Therefore **no finite $(0, 2)$ -graph has chromatic number exactly three.** $\square$

2.5 Scope and gaps

All finite simple graphs in the question are covered, including disconnected graphs by the componentwise argument. No Cayley-graph or cube-like hypothesis is used, and there is no remaining mathematical case in the proof. I have not independently checked whether this argument or the resulting theorem has already appeared in the literature.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	<code>three_chromatic_0_2_graphs</code>
Catalog entry	Three-chromatic (0,2)-graphs
Source corpus	<code>opg</code>
Campaign leg	OpenProblemGarden first pass (<code>attacks_opg</code>)
Source paper / entry	http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort <code>max</code> , mode= <code>pro</code> , flex service tier
Original artifact	<code>attacks_opg/three_chromatic_0_2_graphs/output.md</code>
Independent review	None
Model’s caveats	For finite simple undirected graphs; literature priority has not been independently checked.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/three_chromatic_0_2_g`